

▼ M001 Week 6 — English Worksheet

Topic: Inverted Pyramid · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
set size to 1
while size <= 9:
    [build one size × size square layer]
    move up by 1
    set size to size + 2
```

Each layer gets bigger or smaller? Where is the widest layer — bottom or top?


```
set size to 1
while size <= 9:
    [build one size × size square layer]
    move up by 1
    set size to size + 2
```

List the sizes from bottom to top. How many layers in total?


```
set size to 1
while size <= 5:
    [build one size × size square layer]
    move up by 1
    set size to size + 2
```

How is this different from the code above? How does it look?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to build an **inverted pyramid**: the smallest layer at the bottom, the biggest at the top.

```
set size to 9
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to grow the pyramid from 1 up to 9. Each layer must be **2 blocks longer** than the one below.

```
set size to 1
while size <= 9:
    [build one size × size square layer]
    move up by 1
    set size to size + 1
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Each new layer should sit **on top** of the one below.

```
set size to 1
while size <= 9:
    [build one size × size square layer]
    set size to size + 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build an inverted pyramid that grows from **size 1 up to size 9**. Look from the side and check the shape is right. When you finish, come back here.

Send a photo or video of your pyramid, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

The top layer is ... blocks long.

Each new layer grows by ... because ...

The pyramid has ... layers in total.

The hardest part was ...

Compared to last week's normal pyramid, this one is different because ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk the camera around your inverted pyramid. Talk through it like you are teaching someone who has never seen it. Try to use these words: **inverted, layer, size, grow, stack**.

1. Walk around the pyramid and show that the top is the widest part.
2. Read your **while** loop out loud and say how the size changes each time.
3. Show one bug you hit and how you fixed it.
4. Compare this pyramid to last week's normal pyramid.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.